

Towards Unified Multimodal Misinformation Detection in Social Media: A Benchmark Dataset and Baseline

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Abstract

Detecting deceptive multimodal content on social media has become an increasingly important problem. Two major types of deception dominate: human-crafted misinformation (e.g., rumors and misleading posts) and AI-generated content produced by image synthesis models or vision-language models (VLMs). However, these two types are usually addressed as separate tasks. Consequently, existing models are often specialized for only one type of fake content. In real deployments, however, the fake-content type of an incoming multimodal post is typically unknown, which limits the practicality of such specialized systems. To study this setting, we build OmniFake, a benchmark with 98K samples that combines human-curated misinformation from existing resources with newly created AI-generated examples. To address this new task, we propose Unified Multimodal Fake Content Detection (UMFDet), a framework designed to handle both types of deception. UMFDet builds on a VLM backbone augmented with a Category-aware Mixture-of-Experts (CMoE) adapter to capture category-specific cues. We further introduce an Expert-wise Discriminative Regularization to enforce intra-expert compactness. In addition, cross-modal consistency alignment is proposed to improve the perceptual capability of experts for handling different deception types. Experiments show that UMFDet consistently outperforms competitive specialized baselines across both deception types. The dataset and code will be made publicly available.

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CCS Concepts

• **Do Not Use This Code** → **Generate the Correct Terms for Your Paper**; *Generate the Correct Terms for Your Paper*; Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

Keywords

Multimodal DeepFake Detection, Fake News Detection, Manipulation Recognition

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1 Introduction

The proliferation of multimodal misinformation on social media platforms has emerged as a significant public security concern [12, 23, 32]. In response, the detection of deceptive content like rumors, misleading claims, and fabricated information has attracted considerable research interest in recent years [14, 34, 41]. Mainstream misinformation can be broadly categorized into two types: human-crafted misinformation and AI-synthesized content. Human-crafted misinformation is intentionally created by individuals, often with motives such as gaining followers on social platforms. In contrast, AI-generated content is produced using generative AI tools, including image synthesis models, Large Language Models (LLMs), and Multimodal Large Language Models (MLLMs). The rise of such AI-synthesized misinformation represents a growing challenge in the era of advanced deep learning systems [4, 13].

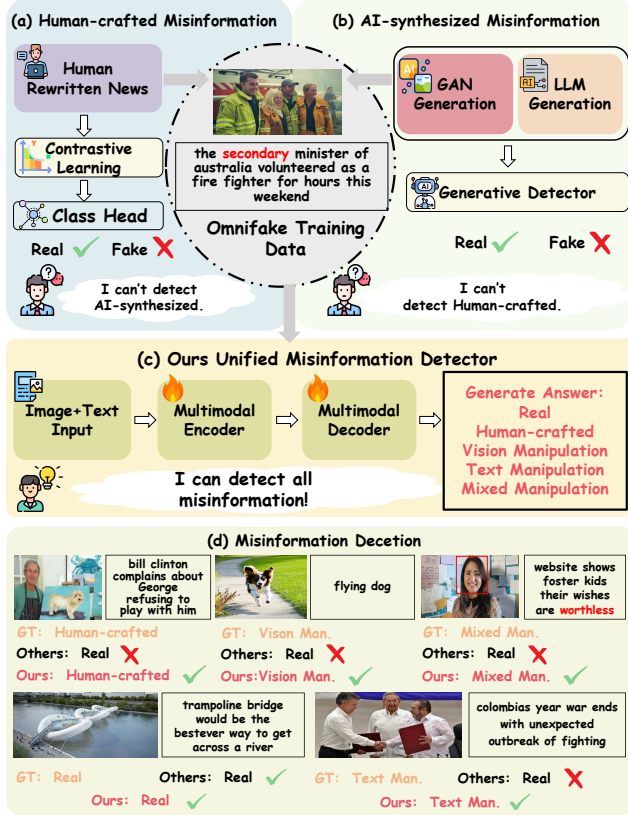


Figure 1: Comparison between conventional detectors and our unified misinformation detector.

Although both human-crafted misinformation and AI-generated content belong to misinformation, they are often studied in isolation. Researchers in natural language processing (NLP) have extensively studied human-crafted deceptive content, such as human-written rumors and misleading information, with a long-standing tradition of fake news detection in social media [8, 16, 21]. On the other hand, the computer vision community has focused primarily on detecting synthetic multimodal media generated by AI models, including deepfakes and other artificially produced visual content [6, 37, 43]. However, in real-world social platforms, both types of misinformation frequently coexist. A straightforward alternative is to maintain two separate detectors, each specialized for one source, and aggregate their predictions at inference time. Nevertheless, such a pipeline implicitly assumes source-specific dispatch and incurs additional computational overhead, making it less desirable for practical deployment.

In response to this challenge, we aim to develop a unified framework capable of simultaneously detecting both human-crafted misinformation and AI-generated fake contents. To support this exploration, we construct a large-scale Omnibus Dataset for Multimodal News Deception (OmniFake). Specifically, we collect source samples from the Fakeddit [26] and VisualNews [26] dataset and select misleading examples to form the category of human-crafted misinformation. For the AI-generated category, we first choose a

subset of real samples from Fakeddit and apply a diverse set of manipulation techniques including face swapping, background replacement, and textual rewriting *etc.*, to simulate various forms of synthetic media. Through this process, We finally construct OmniFake, a dataset comprising 98,592 samples spanning five categories: *Real*, *Human-crafted*, *Vision Manipulation*, *Text Manipulation*, and *Mixed Manipulation*.

The OmniFake dataset presents two distinct yet deceptive forms of misinformation. Beyond the visual discrepancies inherent in human-crafted versus AI-generated news, the textual contents also exhibit considerable heterogeneity (see Fig.2). Motivated by these observations, we move beyond binary classification to treat these as separate categories for fine-grained recognition. Specifically, we propose the Unified Multimodal Fake Content Detection (UMFDet) framework. UMFDet leverages a Multimodal Large Language Model (MLLM) as its backbone and introduces a Category-aware Mixture-of-Experts (MoE) Adapter to tame the MLLM for misinformation detection. This adapter allocates specialized expert modules to identify specific types of deceptive content. To further enhance the model’s ability to accurately discern subtle cues of different types of misinformation, we enforce an Expert-wise Discriminative Regularization (EDR) computed on the routed sets: within each expert subspace, we estimate class prototypes and shrink intra-class variance while enlarging inter-class margins. Furthermore, a cross-modal consistency alignment learning is incorporated to bolster the experts’ perceptual capability towards semantic correlations, effectively distinguishing between real news and human-crafted rumors through global consistency modeling.

In summary, we highlight the contributions of this paper as follows:

- **OmniFake dataset.** We construct OmniFake, a large-scale benchmark that covers two prevalent types of misinformation and enables unified evaluation across them.
- **UMFDet framework.** We propose UMFDet, a unified detection framework that combines a category-aware MoE adapter, expert-wise discriminative regularization, and cross-modal consistency alignment for this setting. Extensive experiments demonstrate its effectiveness.
- **Extensive experimental study on effectiveness and generalization.** We conduct comprehensive evaluations across five datasets. Our proposed model outperforms competing methods in terms of both effectiveness and generalization.

2 Omnibus Dataset for Multimodal News Deception Detection

We build our multimodal fake news dataset on top of r/Fakeddit [26] and VisualNews [20]. r/Fakeddit is a large-scale collection crawled from Reddit, a widely used social news and discussion platform, while VisualNews provides large-scale image–text pairs collected from diverse news media sources. The source corpora contain over one million image–text pairs spanning 22 subreddits and a broad topical spectrum—from political news to everyday posts by Reddit users—as well as professionally curated news content from multiple domains.

To construct an omnibus benchmark for multimodal misinformation, our OmniFake dataset covers three categories: (i) Real posts,

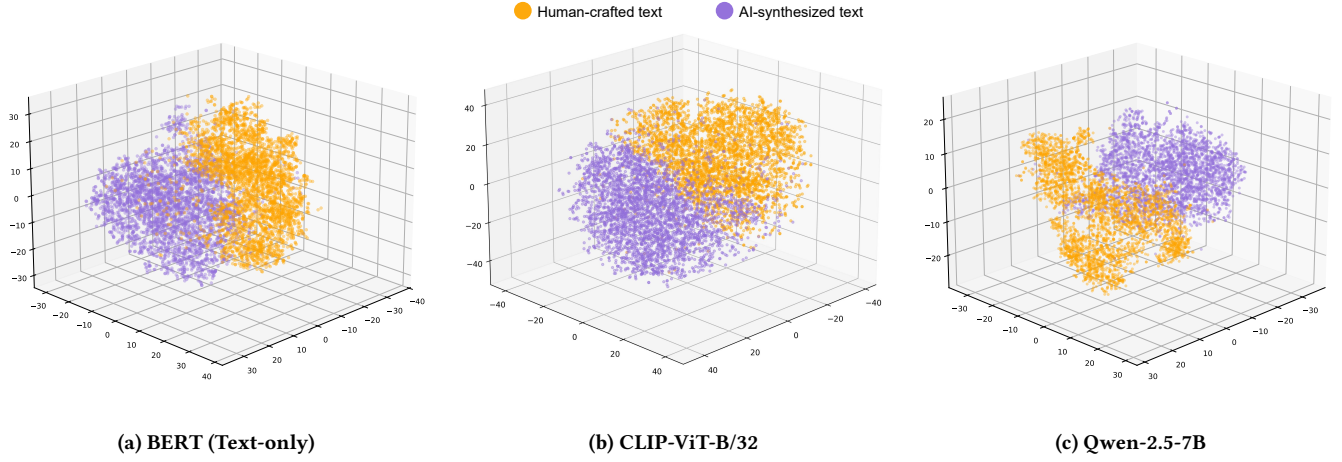


Figure 2: Visualization of textual feature distributions across different backbones. We compare the representation spaces for *Human-crafted text* and *AI-synthesized text*.

Table 1: Comparison with existing multimodal fake news datasets. Modalities: Text (Txt.) and Image (Img.). Category: Real (R), Human-crafted (H), AI-synthesized (A), and Mixed-manipulation (M). Mani.: Manipulated. PIG/E: Part image generation/edit; PTG/E: Part text generation/edit; FTG: Full text generation; FIG: Full image generation. LLM Threat: suitable to evaluate LLM risks. Img./Txt. BBox: availability of image and text bounding-box annotations. Tr.: available Train set.

Datasets	Mod.		Scale	Category	Lang.	Mani.		Img Type		LLM	Img./Txt. BBox	Data Collection	Tr.
	Txt.	Img.				Txt.	Img.	Face	Oth.				
Twitter [3]	☑	☑	18K	R, H	EN	×	×	×	×	×	×	Social platform	☑
Pheme [46]	☑	×	29K	R, H	EN	×	×	×	×	×	×	Social platform	☑
Weibo [27]	☑	☑	11K	R, H	CH	×	×	×	×	×	×	Social platform	☑
NewsClip. [25]	☑	☑	85K	R, A	EN	×	×	×	×	×	×	Social platform	☑
DGM4 [30]	☑	☑	50K	R, A	EN	☑	☑	☑	×	×	☑	Soc.; PIE; PTE	☑
MDSM [42]	☑	☑	441K	R, A	EN	☑	☑	☑	×	×	×	Soc.; PIE; PTE	☑
MMFakeBench [23]	☑	☑	11K	R,H,A	EN	☑	☑	×	☑	☑	×	Soc.; PIG; PIE; FIG; PTE; PTG; FTG	×
OmniFake (Ours)	☑	☑	98K	R,H,A,M	EN	☑	☑	☑	☑	☑	☑	Soc.; PIG; PIE; FIG; PTE; PTG; FTG	☑

(ii) Human-crafted misinformation, and (iii) AI-synthesized content generated with modern image and language tools. Particularly, we select 49,034 posts from r/Fakeddit [26] and VisualNews [20] to form the **Real posts** category in OmniFake. We select 24,726 misleading posts from r/Fakeddit [26] to serve as the **Human-crafted misinformation** in OmniFake. In the following, we elaborate how we generate samples covering diverse manipulation patterns using AI models.

2.1 Synthetic Multimodal News

The proliferation of generative models including LLMs and MLLMs enables heterogeneous pipelines for fabricating persuasive multimodal content, posing new risks to social media [1, 40].

Image Synthesis. We consider two cases of images to construct the synthetic subset: for person-containing images (PCI), we use

state-of-the-art face swapping and face attribute editing models to generate realistic forgeries; for non-person images (NPI), we synthesize data via full-image generation, and image inpainting, yielding diverse, in-the-wild-like samples.

▷ *Face Swap Manipulation (PCI)*. Face swapping has had substantial negative impacts on social media, particularly undermining the credibility of images depicting public figures and politicians. We adopt SimSwap [5], E4S [19] and hyperswap [19] for face swap. Given a source–target pair $\{I_s, I_t\}$, the source image I_s is sampled from source samples, while the target image I_t is randomly selected from CelebA–HQ [45]. We detect faces in both images using the DSFD detector [18], select the highest confidence face pair, and perform landmark based alignment and standardized cropping to confine the swap to the facial region. Conditioned on

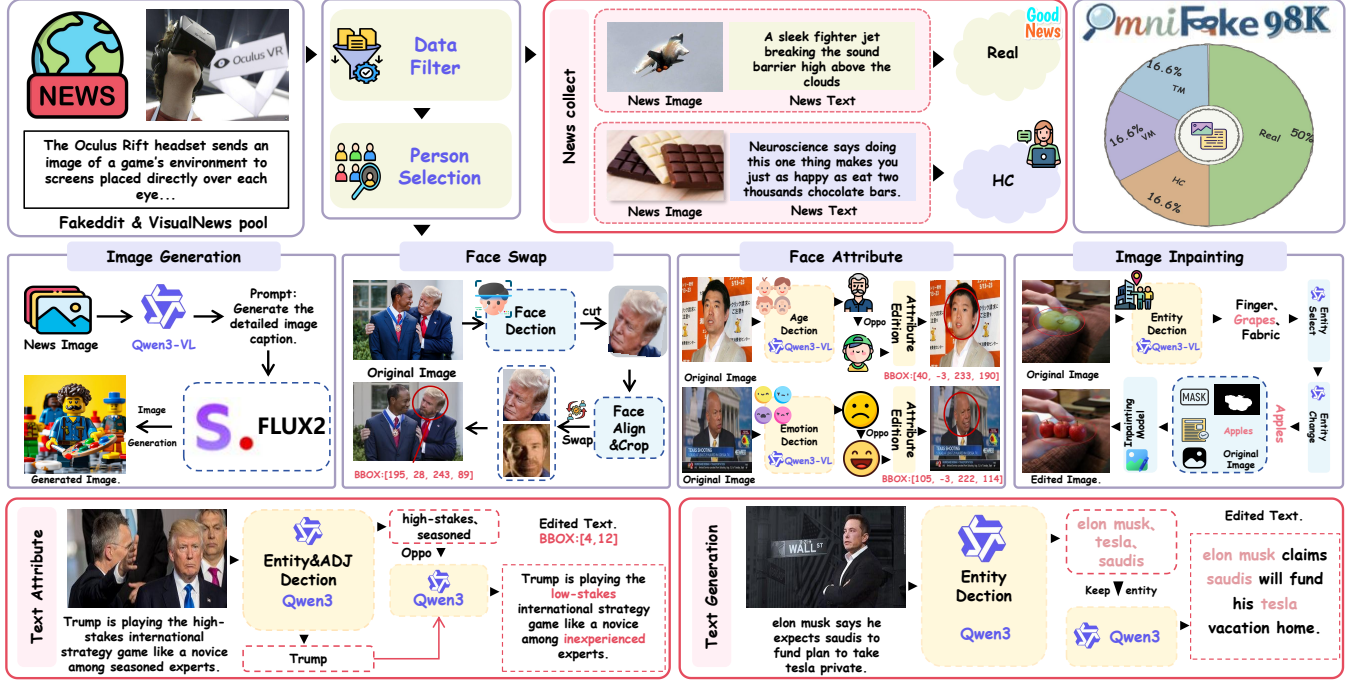


Figure 3: Dataset construction pipeline. We collect authentic and human-crafted posts from the source corpora to construct the *Real* and *Human-crafted* categories. For AI-synthesized misinformation, we generate diverse image-side manipulations, including full-image generation, local inpainting, face swapping, and facial attribute editing, as well as LLM-driven text generation and rewriting with controlled cross-modal semantics.

an identity embedding extracted from I_s , the swapping model renders the source identity onto the target face while preserving the target’s pose, expression, illumination, and non facial context; the generated face is seamlessly blended back into I_t to produce the final swapped image.

▷ *Face Attribute Manipulation (PCI)*. Face attribute editing is another common form of facial manipulation on social media, often used to disseminate misleading emotional content. We employ the HFGI model [33] to edit multiple attributes, including expression changes (e.g., *smile* and *sad*), age, lips, and eyes. Given an input image I , we first detect the most prominent face F using the DSFD detector [18], then perform rotation-based alignment and standardized cropping to obtain the face patch F_c and facial landmarks B . The patch F_c is fed into HFGI [33] with the selected attribute to produce the edited face F_e . For *expression edits*, we use Qwen3-VL [38] to determine whether the original affect is positive or negative and apply reverse emotion editing accordingly. For age edits, we use Qwen3-VL [38] to estimate the original age, while for other facial attributes (e.g., eyes), we feed the aligned face into HFGI [33] and edit along the corresponding latent attribute direction. The edit magnitude is controlled by an editing-strength factor α ; larger α yields stronger effects. The final face is pasted back to the original image guided by landmarks B with seamless blending.

▷ *Full image generation (NPI)*. Rapid advances in text-to-image generation have flooded social media with photorealistic AI images, complicating authenticity checks. To address this, we design a text-driven tampering pipeline: given an original news image, we

first use the large vision-language model Qwen-3-VL [2] to produce a detailed textual description T . The generated description T is then fed into text-to-image generators, including SD-XL [28] and FLUX [11], to synthesize a new news image $\hat{I}_n$ as the tampered sample.

▷ *Object/Background Replacement (NPI)*. Local inpainting replacement is pervasive on social media, where many forged news images alter only a small region to mislead viewers. Given an input image I , we use Qwen-3-VL [2] to produce prompt phrases for the r most salient objects, forming a set $\{n_j\}_{j=1}^r$ (with $r = 3$ by default). We then randomly select one description n_x as the target and feed (n_x, I) into SAM2 [29] to obtain the corresponding object mask M . For local tampering, we adopt an inpainting pipeline based on SDXL Inpainting [28]: a language model converts the source object phrase $p_{src} = n_x$ into a modified phrase p_{mod} by altering its type, color, or attribute while keeping unrelated context unchanged; the inpainting model takes (I, M, p_{mod}) and synthesizes the edited image.

Text Fabrication. On social media, text manipulation can rapidly and easily produce misleading content that distorts facts and fabricates nonexistent news. We consider two manipulation strategies to modify the textual modality. For both strategies, if the text contains *named entities* (e.g., personal or place names), we preserve these key tokens to retain contextual coherence and realism.

▷ *Pure fake text generation.* We take the original news headline T and apply Qwen3 [38] to detect named entities; if present, they

are stored as $\mathcal{E} = \{e_j\}_{j=1}^r$ and kept unchanged. We then feed T_i together with $\mathcal{E}$ into Qwen3 [38], prompting the model to produce a *misleading* fake-news headline that preserves all entity tokens while rewriting the remaining content; if no entities are detected, the *entire* headline is rewritten. This strategy enables comprehensive semantic-level falsification of the text while retaining critical information.

▷ *Key words/phrase Distortion*. We generate misleading content by replacing key adjectives in the headline. Specifically, we apply Qwen3 [38] to the original title T to detect and preserve named entities, yielding the set $\mathcal{E} = \{e_j\}_{j=1}^r$, and extract candidate keywords to form $\mathcal{K} = \{k_j\}_{j=1}^m$. We then feed T , $\mathcal{E}$, and $\mathcal{K}$ into Qwen3 [38] with a prompt that keeps all entities unchanged while either randomly selecting 2–3 items from $\mathcal{K}$ or choosing short phrases in the text for controlled replacements—using antonyms for adjectives—thus producing the partly rewritten headline T . Throughout this process, we log each replaced keyword k_x or phrase p_x and its position P to facilitate subsequent detection.

2.2 Dataset Statistics

As shown in Figure 3, with the above pipeline, we curate OmniFake, a multimodal fake-news corpus of 98,592 image–text pairs. Unlike prior sets that separate human-crafted misinformation from AI-synthesized media and mostly adopt binary labels, OmniFake natively supports both *binary* detection (Real / Fake) and *five-class* detection (*Real*, *Human-crafted*, *Vision Manipulation*, *Text Manipulation*, and *Mixed Manipulation*). Notably, our taxonomy explicitly accounts for hybrid deception cases involving both image-side and text-side interventions. Specifically, the *Mixed Manipulation* category comprises eight manipulation patterns, formed by combining four image-side manipulation types with two text-side manipulation types. Beyond image-side edits, we generate text-side variants with LLMs to produce fluent yet image-misaligned headlines—an underexplored but realistic challenge [did not understand](#). The corpus in OminiFake spans multiple domains (politics, health, people, society, *etc.*), promoting broad coverage and cross-domain generalization. It also covers a wide manipulation spectrum, including full-image generation, local inpainting/region edits, face swap and attribute edits, and text rewriting (keyword/phrase substitutions). OmniFake is an early attempt to build large-scale multimodal benchmark to jointly offer multi-class labels and such breadth of manipulations for unified multimodal manipulation detection. [highlight balance](#)

2.3 Quality Evaluation

We conduct a human audit on a randomly subset of OmniFake. As shown in the top table (600 samples), human observers (8 master students) often struggle to distinguish AI-synthesized samples from genuine content, indicating the dataset is sufficiently realistic to deceive humans. Regarding AI artifacts, the bottom table (100 tests) shows that [over 77% of generated images](#) are taken as real ones, while also receiving high quality scores, further validating the visual quality of our dataset. [Too rough](#).

Table 2: Human evaluation results on a sampled subset ($N = 600$). Top: overall five-class classification performance. Bottom: visual realism evaluation of AI-synthesized images, focusing on generator-specific artifacts.

Evaluator	ACC (%)	Pre (%)	Recall (%)	F1 (%)
Human	40.00	39.30	40.14	33.46

Image Samples	Image Quality Evaluation Metric	Score
50 Real + 50 Fake	Recognition Accuracy (50% indicates random guessing) ↑	44.29%
	Quality Score (−1: fake; 1: real) ↑	0.434
50 Fake	Mis-recognition Percentage ↑	77.43%
	Quality Score (−1: fake; 1: real) ↑	0.549

3 UMFDet Model

3.1 Architecture

As illustrated in Figure 6, we take Florence-2 (0.7B) [36] as the backbone to balance efficiency and effectiveness. We adapt the MLLM to multimodal misinformation detection with a Category-aware Mixture of Experts (CMoE). Following the MLLM paradigm, we formulate misinformation detection as a question-answering task, where each news sample is converted into a task-oriented prompt composed of the image and its corresponding title. The image is first encoded by a vision encoder to obtain patch embeddings, while the textual prompt is processed through an embedding layer to derive text representations. The multimodal embeddings are then fused and fed into the multimodal encoder, followed by the CMoE module and a language decoder for answer generation. To enhance expert specialization, we introduce an Expert-wise Discriminative Regularization (EDR) applied to the CMoE outputs, which encourages discriminative learning across experts. Furthermore, we incorporate cross-modal consistency alignment learning to strengthen the model’s capability in distinguishing between real, human-crafted misinformation across different experts. The entire network is jointly optimized with the proposed EDR loss, the language modeling loss, and the cross-modal consistency alignment loss.

Image Embeddings. Given input image I , we use DaViT [10] as our visual encoder, generating flattened visual token embeddings $E_v \in \mathbb{R}^{N_v \times H}$, where N_v represents the number of tokens and H is the dimension of each token.

3.2 Instruction Design and Embedding

We build the instruction template as following:

$$T = [P_{\text{task}}; P_{\text{opt}}(C); P_{\text{que}}; P_{\text{resp}}], \quad (1)$$

where P_{task} specifies the goal, $P_{\text{opt}}(C)$ lists the category set and brief definitions, P_{que} define the question, and P_{resp} declares the expected answer format. The textual instruction is designed in the following format:

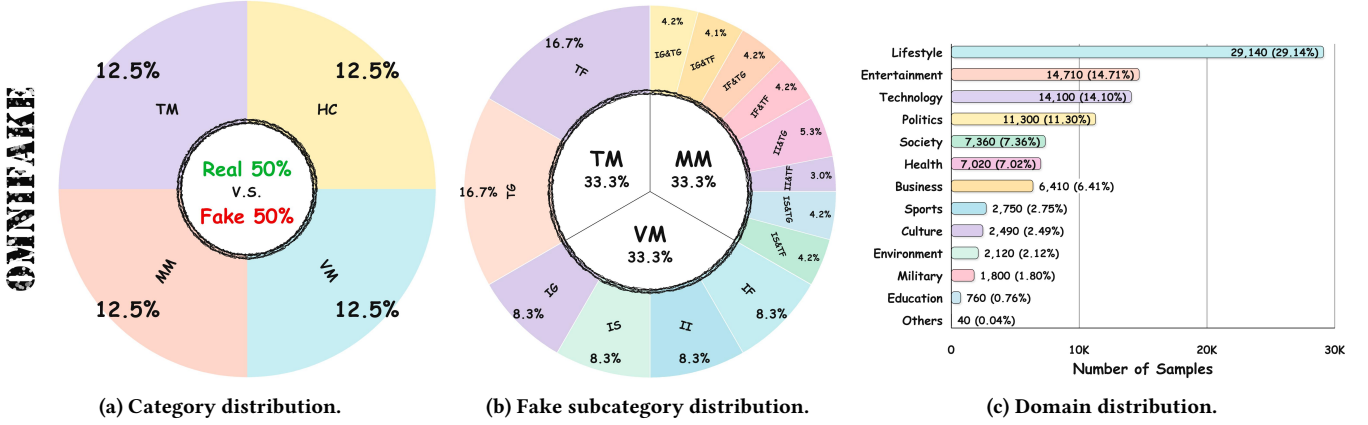


Figure 5: Overview of the OmniFake dataset. Top: Statistical distributions of the five target categories, fine-grained fake manipulation types, and news domains, demonstrating the balanced category design, diverse manipulation coverage, and broad topical distribution. Bottom: Representative image-text pairs from the *Real*, *Human-crafted*, *Vision Manipulation*, *Text Manipulation*, and *Mixed Manipulation* categories.

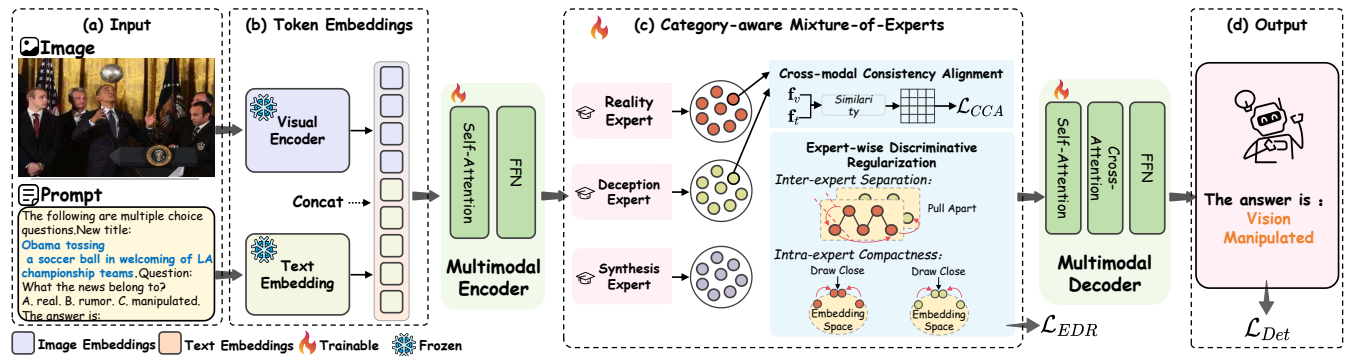


Figure 6: We propose a multi-modal fake news detection framework based on a Mixture-of-Experts (MoE) architecture UMFDet.

P_{task} : Given the news image and news title (I, T), analyze which category the news belongs to.

$P_{\text{opt}}(C)$: Define the three categories and guide the model.

P_{que} : Define a question and have the model think and answer.

P_{resp} : Finally, encapsulate the correct category.

After tokenization and embedding, we obtain $E_t \in \mathbb{R}^{N_t \times H}$ for prompt representation.

3.3 Category-aware Mixture-of-Expert

Our Category-aware MoE (CMoE) uses three [we now have five categories](#) experts, each aligns with one information category (real, human-crafted, AI-synthesized). Given textual embeddings $E_t \in \mathbb{R}^{N_t \times H}$ and visual embeddings $E_v \in \mathbb{R}^{N_v \times H}$, We fuse the concatenated multimodal inputs using a language encoder:

$$X = L_{\text{enc}}([E_v; E_t]) \in \mathbb{R}^{(N_v+N_t) \times H}, \quad (2)$$

Subsequently, to capture category-specific features, we introduce a Category-aware Mixture-of-Experts (MoE) module composed of a router network and 3 ! experts. In particular, we define three experts as: Reality Expert $\mathcal{E}_R$, Deception Expert $\mathcal{E}_D$, and Synthesis Expert $\mathcal{E}_S$, each specialized for a distinct information category. Given a multimodal news representation X , the experts yield encodings $[\mathcal{E}_R(X), \mathcal{E}_D(X), \mathcal{E}_S(X)]$. Formally, each expert is a feed-forward network.

$$\begin{cases} \mathcal{E}_e(X) = \text{Dropout}(y) W_{\text{out}} + b_{\text{out}}, \\ y = u \odot v, \\ u = \text{SiLU}(XW_a + b_a), \\ v = \text{sigmoid}(XW_b + b_b), \end{cases} \quad (3)$$

where all W and b are learnable parameters. The router network, implemented as a linear layer with softmax activation, then selects the optimal expert index e as follows:

$$\begin{cases} e = \arg \max_{e \in \{R,D,S\}} \{\mathcal{W}_R, \mathcal{W}_D, \mathcal{W}_S\}, \\ [\mathcal{W}_R, \mathcal{W}_D, \mathcal{W}_S] = \text{Softmax}(WX + b), \end{cases} \quad (4)$$

Finally, we select the $\mathcal{E}_e(X)$ and feed it into the language decoder for answer generation. Language modeling loss is employed for training:

$$\mathcal{L}_{\text{Det}} = - \sum_{t=1}^{|Y|} \log p_\theta(y_t | y_{<t}, \mathcal{E}_e(X)), \quad (5)$$

3.4 Expert-wise Discriminative Regularization

For each expert, we expect it to specialize in a particular type of manipulation. To this end, we add a simple batch-wise regularizer on the CMoE outputs.

Let $\mathcal{S}_r = \{X_i\}_{i=1}^{|\mathcal{S}_r|}$ denote the set of samples routed to the real expert in the current mini-batch. Similarly, the sample sets corresponding to the deception expert and the synthesis expert are denoted as $\mathcal{S}_d$ and $\mathcal{S}_s$, respectively. To ease the constraint and stabilize the optimization, we dynamically select the two experts that receive the largest number of samples within each mini-batch. Formally, the selected sets are defined as:

$$\{S_1, S_2\} = \text{Top-2}(\{\mathcal{S}_r, \mathcal{S}_d, \mathcal{S}_s\}, \text{key} = |S|), \quad (6)$$

Within each expert subspace, we define its prototype and variance as:

$$\mu_k = \frac{1}{|\mathcal{S}_k|} \sum_{i=0}^{|\mathcal{S}_k|} X_i, \quad \sigma_k^2 = \frac{1}{|\mathcal{S}_k|} \sum_{i=0}^{|\mathcal{S}_k|} \|f_i - \mu_k\|_2^2, \quad (7)$$

$$d_{1,2} = \|\mu_1 - \mu_2\|_2^2, \quad (8)$$

where $k \in \{1, 2\}$. Then, a clustering loss is defined to minimize within-expert variance and maximize inter-expert distance:

$$\mathcal{L}_{\text{EDR}} = \frac{\sigma_1^2 + \sigma_2^2}{d_{1,2} + \varepsilon}, \quad (9)$$

where $\varepsilon > 0$ is a small number for computation stability. EDR term encourages the samples with each set more compact, while pushes the two expert prototypes further apart.

3.5 Cross-modal Consistency Alignment

In our practice, we found that the trained models still struggle to distinguish between authentic news and sophisticated human-crafted misinformation. Notably, the multimodal misinformation crafted by humans often repurposes images from unrelated sources, pairing them with misleading or out-of-context text. As a result, the semantic alignment between image and text in such deceptive content is typically weaker than that in authentic posts. Leveraging this prior, we introduce a Cross-modal Consistency Alignment (CCA) mechanism within the CMoE framework, which ensures that samples routed to different experts adhere to the underlying multimodal correlation prior.

Mathematically, let $X = [X_v; X_t]$ denote a sample from real/deception expert set, where X_v and X_t are the visual and textual splits of the multimodal representation sequence X . Subsequently, we define the target cross-modal alignment values as $y_{\text{sim}} = 1$ for real expert set and $y_{\text{sim}} = -1$ for the deception expert set ($\mathcal{C}_{hc}$). The loss is calculated using MSE over the respective subsets:

$$\mathcal{L}_{\text{CCA}} = \sum_{X \in \mathcal{S}^r} \|\text{sim}(X_v, X_t) - 1\|^2 + \sum_{X \in \mathcal{S}^d} \|\text{sim}(X_v, X_t) - (-1)\|^2, \quad (10)$$

where $\text{sim}(\cdot, \cdot)$ is the cosine similarity function.

Finally, the total training objective of our UMFDet is a joint optimization of the language modeling loss $\mathcal{L}_{\text{Det}}$, the expert-wise discriminative regularization $\mathcal{L}_{\text{EDR}}$, and the alignment loss $\mathcal{L}_{\text{CCA}}$:

$$\mathcal{L} = \mathcal{L}_{\text{Det}} + \lambda(\mathcal{L}_{\text{EDR}} + \mathcal{L}_{\text{CCA}}). \quad (11)$$

4 Experiment

Experiment Setup. Comprehensive details regarding the experimental configurations and an visualization of the expert routing mechanisms are provided in the Supplementary Material.

4.1 Comparison with State-of-the-Art Methods

Quantitative Results. To evaluate the performance of the proposed framework, we conduct a comprehensive comparison with a wide range of state-of-the-art methods on the OmniFake dataset. As summarized in Table 3, UMFDet substantially outperforms all baseline models under the unified five-class classification setting. Specifically, it achieves an average precision of 82.56%, an average recall of 82.53%, and an average F1-score of 82.23% across the five categories.

[adjust the order of figure and tables](#). UMFDet demonstrates clear advantages across several challenging subsets. In the Real category, UMFDet attains an F1-score of 89.89%, exceeding the best baseline by 4.49%. Notably, for the Human-crafted class our method achieves a precision of 92.17%, a recall of 82.22%, and an F1-score

Table 3: Comparison of multi-modal learning methods on OmniFake. AVG refers to the average performance across 5 classes.

Setting	Method	Venue	Real			Human-crafted			Vision Manipulation			Text Manipulation			Mixed Manipulation			AVG 5-class		
			Pre	Recall	F1	Pre	Recall	F1	Pre	Recall	F1	Pre	Recall	F1	Pre	Recall	F1	Pre	Recall	F1
Zero-shot	Gemini-2.5 [7]	-	42.86	42.66	42.76	55.00	47.50	50.98	54.55	56.55	55.53	42.86	58.00	50.30	19.99	47.35	14.99	43.05	50.41	42.91
	Qwen2.5-VL-72B [2]	-	44.81	49.87	47.20	36.08	40.38	38.11	45.61	46.50	46.05	41.54	44.91	43.16	47.00	42.64	44.71	43.01	44.86	43.85
	DeepSeek-VL2-27B [35]	-	41.21	46.18	43.55	46.67	43.18	44.86	48.11	50.12	49.09	38.95	37.01	37.95	41.29	47.56	44.20	43.24	44.81	43.93
	GPT-4o [1]	-	60.00	59.24	60.59	43.70	37.58	40.41	41.34	45.22	42.78	40.51	51.59	45.36	44.58	31.21	35.32	46.03	44.97	44.89
OmniFake	FKA_OWL [22]	MM'24	78.71	93.47	85.40	88.18	72.08	78.32	65.37	56.09	60.38	49.91	21.67	30.22	61.35	69.97	65.38	68.70	62.65	64.15
	HAMMER [30]	CVPR'24	82.98	80.93	81.94	74.98	73.67	74.32	71.21	73.86	72.51	52.12	51.30	51.70	74.17	72.97	73.57	71.09	70.55	70.81
	MIMOE [24]	WWW'25	81.33	86.51	83.84	80.06	66.31	72.54	69.12	78.02	73.30	56.81	51.45	53.45	73.56	65.86	69.50	72.58	69.83	70.93
	GLPN [15]	ACL'25	59.81	95.45	73.54	69.80	57.79	63.23	51.15	12.66	20.30	22.22	0.81	1.57	54.75	27.60	36.70	51.55	38.86	39.07
	Ours	-	87.04	92.94	89.89	92.17	82.22	86.92	79.49	63.23	70.43	71.95	61.85	66.52	68.68	81.17	74.40	82.56	82.53	82.23

of 86.92%, outperforming all competing methods. For Text Manipulation, UMFDet achieves a precision of 71.95%, a recall of 61.85%, and an F1-score of 66.52%, improving upon the strongest baseline by 13.07% in F1-score. It also obtains the best recall of 81.17% and the best F1-score of 74.40% on Mixed Manipulation. These results collectively demonstrate that UMFDet provides a robust and practical solution for unified multimodal misinformation detection, outperforming established multimodal baselines across diverse manipulation settings.

Performance on MMFakeBench. We evaluate the performance of UMFDet on the mixed-source MMFakeBench dataset [23]. As illustrated in Table 4, UMFDet achieves the best overall performance on both the validation and test sets. On the validation set, our model attains an F1-score of 61.74% and an accuracy of 61.17%. On the test set, it reaches an F1-score of 61.70%, a precision of 70.94%, a recall of 61.12%, and an accuracy of 61.14%, consistently outperforming all reported baselines. In particular, it also outperforms representative MLLM baselines across different parameter scales, including the best 7B model under the Standard setting (52.5% F1) and the 13B model enhanced by the MMD-Agent framework (50.2% F1). These results highlight the robustness of our approach in detecting mixed-source multimodal misinformation.

Performance on DGM4. Furthermore, we evaluate the effectiveness of UMFDet on the DGM4 dataset [30], a benchmark dataset for multimodal media manipulation detection. As illustrated in Table 5, UMFDet achieves the best overall performance, reaching an accuracy of 83.10% and an F1-score of 83.20%, consistently outperforming all state-of-the-art baselines. Notably, on the Real subset, our model achieves the highest accuracy of 86.88% and the highest recall of 86.88%, exceeding HAMMER by 6.25% in accuracy and 4.05% in recall, respectively. Regarding the AI-synthesized category, UMFDet delivers the strongest precision of 92.85%, outperforming HAMMER by a substantial margin of 8.57% (92.85% vs. 84.28%). [Any more recent comparison works?](#)

Performance on PHEME. We further evaluate the efficacy of UMFDet on the PHEME dataset for rumor detection. As shown in Figure 7a, UMFDet achieves a Macro-F1 score of 74.3%, outperforming both zero-shot LLMs and specialized early-detection methods. Compared with the best-performing LLM-based variant, EARD w/ ChatGPT (64.6%), UMFDet yields an absolute improvement of 9.7%. Notably, UMFDet attains the lowest Early Rate (ER) of 18.7%, improving upon the prior state-of-the-art method, HEARD (26.6%), by 7.9%.

Generalization Evaluation. Figure 7b reports zero-shot cross-domain performance on DGM4. UMFDet achieves the best results across all reported metrics, including accuracy, precision, recall, and F1-score. In particular, it reaches an F1-score of 54.86%, exceeding the strongest baseline, FKA-OWL, by 16.86%. These results demonstrate robust zero-shot transfer to diverse multimodal news domains. Figure 7c presents zero-shot results on the FineFake benchmark [44]. UMFDet achieves the best overall performance, attaining an accuracy of 53.64% and an F1-score of 55.10%. In particular, UMFDet improves the F1-score over FKA-OWL and HAMMER by 7.13 and 11.39%, respectively, further demonstrating its robustness under distribution shifts.

4.2 Ablation Study

Contributions of CMoE, EDR, and CCA. Table 6(a) evaluates the contribution of each component. [make abbr. consistent](#). CMoE alone achieves 76.35% ACC and 75.99% F1. Adding Expert-wise Discriminative Regularization (EDR) improves the results to 79.71% ACC and 78.03% F1, validating its role in enhancing expert discrimination. Incorporating Cross-modal Consistency Alignment (CCA) further yields the best performance, reaching 82.53% ACC, 82.56% precision, 82.53% recall, and 82.23% F1. These results demonstrate the complementary effects of expert specialization, discriminative regularization, and cross-modal alignment.

Effect of the Number of MoE Layers. As shown in Table 6(b), one CMoE layer achieves the best performance, improving ACC and F1 from 74.12% and 74.63% to 82.53% and 82.23%, respectively. Adding a second layer reduces ACC and F1 to 80.31% and 80.06%, suggesting that deeper expert routing introduces unnecessary complexity. We therefore adopt a single CMoE layer.

Effect of the Hyper-parameter λ . Table 6(c) reports the effect of the loss weight λ . Performance improves from 79.26% ACC and 80.12% F1 at $\lambda = 0.01$ to the best results of 82.53% ACC and 82.23% F1 at $\lambda = 0.05$. Increasing λ to 0.1 decreases ACC and F1 to 77.20% and 76.45%, indicating over-regularization. We thus set $\lambda = 0.05$ in all experiments.

Performance with a larger backbone. As shown in Table 7(a), UMFDet with Qwen2.5-VL-3B improves precision and F1 from 88.20% and 79.34% to 88.41% and 82.88%, respectively. We use Florence-0.7B as the default backbone for a better effectiveness–efficiency trade-off.

Table 4: Comparison of zero-shot binary detection performance on the MMFakeBench validation and test sets. We compare our proposed REFORM with baseline models using “Standard” and “MMD-Agent” prompting paradigms as defined in the MMFakeBench paper. Baseline results are cited from the original MMFakeBench publication.

Model	Language	Prompt	Validation (1000)				Test (10000)			
Name	Model	Method	F1	Precision	Recall	ACC	F1	Precision	Recall	ACC
LVLMS with 7B Parameter										
InstructBLIP [9]	Vicuna-7B [9]	Standard	14.7	30.8	13.2	8.1	16.1	40.5	14.2	8.8
Qwen-VL [2]	Qwen-7B [2]	Standard	43.6	50.6	44.9	60.3	44.0	51.6	45.2	60.5
PandaGPT [31]	Vicuna-7B [9]	Standard	24.6	60.6	50.5	30.9	24.1	61.7	50.4	30.6
mPLUG-Owl2 [39]	LLaMA2-7B [17]	Standard	47.2	64.9	52.3	70.6	48.7	71.1	53.3	71.4
LLaVA-1.6 [17]	Vicuna-7B [9]	Standard	48.1	48.2	48.5	59.5	52.5	53.0	52.6	62.5
LVLMS with 13B Parameter										
InstructBLIP [9]	Vicuna-13B [9]	Standard	41.1	35.0	49.9	69.9	41.1	35.0	49.9	69.8
		MMD-Agent	51.3	53.4	54.0	53.1	47.9	50.1	50.1	49.9
		Standard	41.1	35.0	50.0	69.7	42.3	57.3	50.1	69.5
LLaVA-1.6 [17]	Vicuna-13B [9]	MMD-Agent	51.8	66.7	54.6	71.4	50.2	67.3	53.9	71.3
Trian on omnifake										
HAMMER [30]	-	-	56.68	69.54	57.62	57.64	57.29	71.32	58.11	58.21
Ours	Florence2-0.7B [36]	Ours (Sec.3.2)	61.74	70.16	61.14	61.17	61.70	70.94	61.12	61.14

Table 5: Comparison of multimodal learning methods on DGM4. The best results in each group are highlighted in bold. “ALL” denotes performance over the entire dataset.

Method	Venue	AVG				Real				AI-synthesized			
		ACC	Pre	Recall	F1	ACC	Pre	Recall	F1	ACC	Pre	Recall	F1
Qwen-2.5-VL-72B [2]	arXiv’25	60.06	57.48	57.81	57.64	60.46	42.71	50.92	46.46	60.12	72.26	64.71	68.28
GPT-4o [1]	arXiv’24	62.14	59.20	58.45	58.82	62.75	45.03	52.86	48.56	61.93	74.10	66.48	69.55
DeepSeek-VL2-27B [35]	arXiv’25	57.23	62.51	57.12	59.69	59.54	57.17	66.14	61.33	43.21	67.82	48.03	56.23
Gemini-2.5 [7]	arXiv’25	67.20	64.05	63.30	63.67	67.90	48.60	57.10	52.51	66.85	78.90	71.20	74.85
MIMOE [24]	WWW’25	62.11	58.74	59.32	58.84	62.11	44.02	50.98	47.25	62.11	73.45	67.65	70.43
FKA-OWL [22]	MM’24	76.37	76.91	78.58	79.30	69.62	74.09	70.47	82.82	76.14	93.92	72.67	81.94
HAMMER [30]	CVPR’24	80.62	81.27	85.13	83.16	80.63	66.87	82.83	74.00	80.63	84.28	89.53	86.83
UMFDet (Ours)	Ours	83.10	81.35	84.62	83.20	86.88	69.21	86.88	77.14	81.18	92.85	81.18	86.62

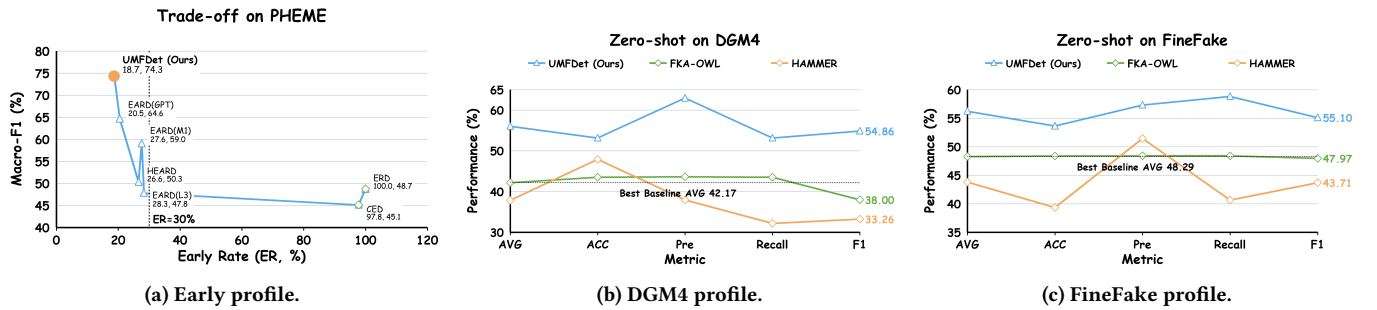


Figure 7: Performance profiles of UMFDet under different evaluation settings.

Number of Experts. Table 7(b) shows that three experts achieve the best overall performance, reaching 82.56% precision, 82.53% recall, and 82.23% F1. Both two- and four-expert configurations perform worse, supporting our default design.

Open-world Test. We additionally collect 100 real-time news samples for evaluation. As shown in Table 7(c), UMFDet achieves the best accuracy (52.00%) and F1-score (51.04%), outperforming HAMMER and FKA-OWL in this challenging open-world setting. [Merge table 6 and 7](#)

Table 6: Ablation study including (a) component-wise analysis, (b) number of MoE layers and (c) discussion of λ .

Components			ACC	Prec.	Rec.	F1
MoE	Regu	CCA				
☐			76.35	76.71	76.35	75.99
☐	☐		79.71	80.33	76.58	78.03
☐	☐	☐	82.53	82.56	82.53	82.23

(a) Component ablation

Layers	ACC	Prec.	Rec.	F1
0	74.12	76.05	74.12	74.63
1	82.53	82.56	82.53	82.23
2	80.31	80.62	79.51	80.06

(b) Number of MoE layers

Weight	ACC	Prec.	Rec.	F1
0.01	79.26	76.92	81.33	80.12
0.05	82.53	82.56	82.53	82.23
0.1	77.20	76.97	77.20	76.45

(c) Hyper-parameter λ

Table 7: Additional analyses of (a) larger backbone validation, (b) different numbers of experts, and (c) open-world generalization.

Method & Backbone	Pre.	Rec.	F1
Qwen2.5-VL-3B (Stand.)	88.20	77.13	79.34
UMFDet (Qwen2.5-VL-3B)	88.41	81.25	82.88
UMFDet (Florence-0.7B)	82.56	82.53	82.23

(a) Larger backbone validation

Experts	Pre.	Rec.	F1
2	76.64	76.59	76.56
3 (Ours)	82.56	82.53	82.23
4	79.63	80.21	79.92

(b) Number of experts

Metric	Ours	HAMMER	FKA-OWL
ACC	52.00	48.00	47.00
F1	51.04	46.26	41.37

(c) Open-world test

5 Conclusion

In this work, we tackle the challenge of detecting diverse multi-modal misinformation spanning both human-crafted and AI-generated manipulations. We introduce OmniFake, a 98K-sample benchmark covering five categories: *Real*, *Human-crafted*, *Vision Manipulation*, *Text Manipulation*, and *Mixed Manipulation*. We further propose UMFDet, a unified detection framework that combines a VLM backbone, a Category-aware MoE adapter, an expert-wise discriminative regularization mechanism, and cross-modal consistency alignment to enhance the experts' capability of modeling category-specific cues and image-text semantic correlations. Through extensive experiments, we demonstrate that UMFDet achieves robust and generalizable performance, surpassing specialized baselines and providing a practical solution for unified multimodal misinformation detection.

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